CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 3: Compound Interest (Using Formula)

Every formula and result of the chapter on one sheet

1	Amount, compounded yearly	$A = P\left(1 + \frac{r}{100}\right)^n$
2	Compound interest	$A - P$
3	Different rates	$A = P\left(1 + \frac{r_1}{100}\right)\left(1 + \frac{r_2}{100}\right)\ldots$
4	Compounded half-yearly	$A = P\left(1 + \frac{r}{200}\right)^{2n}$
5	Compounded quarterly	$A = P\left(1 + \frac{r}{400}\right)^{4n}$
6	Time $n + \frac{1}{k}$ years	$A = P\left(1 + \frac{r}{100}\right)^n\left(1 + \frac{r}{100k}\right)$
7	Principal	$P = A \div \left(1 + \frac{r}{100}\right)^n$
8	C.I. $-$ S.I., 2 years	$P\left(\frac{r}{100}\right)^2$
9	C.I. $-$ S.I., 3 years	$P\left(\frac{r}{100}\right)^2\left(3 + \frac{r}{100}\right)$
10	Growth	$V = V_0\left(1 + \frac{r}{100}\right)^n$
11	Depreciation	$V = V_0\left(1 - \frac{r}{100}\right)^n$
12	Value n years ago	$V_0 \div \left(1 + \frac{r}{100}\right)^n$